MILITARY SPECIFICATION

TRUCK WRECKER: 10 TON, 4X4, M553

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high mobility (GOER type) general and special purpose, 10 ton wrecker crane, diesel engine driven, two or four-wheel optional-drive military truck. This vehicle shall operate cross-country, swim inland waterways, and be air transportable.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bid or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

VV-F-800	-	Fuel Oil, Diesel.

Military

MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank.
MIL-R-3065	-	Rubber, Fabricated Products.
MIL-A-8421	-	Air Transportability Requirements, General Specification for.
MIL-P-17802	-	Padlocks, and Padlock Sets, Pin Tumbler Mechanism, Key Operated.
MIL-A-46153	-	Antifreeze, Ethylene Glycol, Inhibited, Heavy-Duty, Single Package.
MIL-V-62038	-	Vehicles, Wheeled: Preparation for Shipment and Storage of.
MIL-T-62129	-	Tire, Pneumatic: For Truck, Logistical GOER Type, (Tubeless).
MIL-E-62133	-	Engine, Diesel: 6 Cylinder, In-Line, 213 H.P. Turbo-Supercharged.
MIL-T-62134	-	Transmission, Power Shift, Commercial Modified.

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STANDARDS

Federal

FED. STD. NO. 595 - Colors.

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-130	-	Identification Marking of U. S. Military Property.
MIL-STD-193	-	Painting Procedures, Tactical Vehicles, (Tracked and Wheeled).
MIL-STD-210	-	Climatic Extremes for Military Equipment.
MIL-STD-417	-	Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests).
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	-	Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-882	-	System Safety Programs for Systems and Associated Subsystems and Equipment: Requirements for.
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	-	Noise Limits for Army Materiel
MS-51326-1	-	Heater, Vehicular Compartment: Liquid Coolant, 30,000 BTU/HR.
MS-53063-5	-	Indicator, Air Cleaner, Intake, Restriction Mechanical Type.

DRAWINGS

11614318	-	Truck, Wrecker: 10 Ton, 4x4, M553.
11614938	-	Rear Canopy Kit.
11614953	-	Mounting Kit, Mach Gun, Cal 50.
11614987	-	Artic Crane Cab Kit.
11614988	-	Artic Kit.
11624639	-	Filter Kit.
11632925	-	Shipping and Stowage Kit.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction vehicle. When specified (see 6.2), a complete preproduction vehicle(s) to be furnished under this specification shall be produced prior to the manufacture or fabrication of vehicles in quantity. The vehicle(s) when complete shall be submitted to the Government for examination and test to determine conformance to the requirements of this specification. Vehicle(s) submitted by the contractor shall be fully representative of vehicles proposed to be supplied by the vehicle manufacturer from production facilities and tooling under the supply contract.

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3.2 Initial production vehicle(s). The vehicle(s) selected shall be fully representative of vehicles proposed to be furnished under the contract and shall be submitted to the Government for inspection to determine conformance to the requirements of this specification.

3.2.1 Reliability (specified). During the first 12,500 miles of operation, the vehicle shall have a Mean Miles Between Mission Failure (MMBMF) of 925 miles. A mission failure is defined as a malfunction which:

(a) Will immobilize the vehicle, produce a critical or catastrophic safety hazard as defined by MIL-STD-882.
(b) Cannot be corrected within one hour (including diagnostic time) by the crew using On Equipment Material (OEM).

3.2.1.1 Vehicle mission. Reliability operation shall be conducted in the ratio of mileage and requirements of the mission statement (see 6.5).

3.2.2 Durability (minimum). During the first 12,500 miles of vehicle operation, the engine and transmission shall have a combined probability of 70% of not requiring replacement.

3.2.3 Maintainability. When maintenance procedures specified in the technical manuals are followed and the vehicle is operated as defined in the mission statement, the demonstrated maintainability shall be as follows:

(a) Mean Operator Preventive Maintenance Time shall not exceed 1.40 manhours per mission. This time shall not be included in organizational preventive maintenance time.
(b) Maximum Operator Corrective Maintenance Time shall not exceed 1.0 manhours per mission without being classified as a mission failure.
(c) The ratio of total corrective and organizational preventive maintenance manhours to operating hours shall not exceed .50.
(d) The ratio of total organizational preventive maintenance manhours to total operating hours shall not exceed .15.
(e) The ratio of total corrective maintenance manhours to operating hours shall not exceed .35.
(f) Mean manhours to perform a corrective maintenance action (MTTR) shall not exceed 4.67.
(g) The Mean Time Between Corrective Maintenance Actions (MTBF) shall not be less than 13.3 operating hours.
(h) Inherent Availability (Ai) shall be not less than .85.

$$Ai = \frac{MTBF}{MTBF + MTTR}$$

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NOTE: In the above formula, MTBF is expressed in engine hours and MTTR is expressed in clock hours. Where specific hours are not available, appropriate conversion factors (i. e., to change from manhours of maintenance to clock hours of down time) should be used.

3.3 Materials.

3.3.1 Qualified products list (QPL). The contractor shall be responsible for using all assemblies and parts necessary in vehicle construction that are listed as qualified products on the QPL and those approved for inclusion on the QPL. Assemblies and parts determined as QPL items appear on the Engineering List of Specifications cited in 3.4. The contractors manufacturing inspection records shall specifically list all QPL items by part or drawing number(s), name of supplier(s), and number and date of the QPL. When assemblies and parts are approved as qualified products but not as yet listed on the QPL, the contractor shall list the products by number and date of the approval document with name of supplier(s).

3.3.2 Ozone resistant products. Rubber products procured under requirements of MIL-R-3065 are to be ozone resistant to the climatic environment specified herein and the degree specified by the requirements of MIL-STD-417. Rubber components controlled by detail specifications shall be ozone resistant only when required by the specification referenced on the drawing.

3.4 Construction. The vehicle, components, subassemblies, and assemblies shall be fabricated and assembled into a complete vehicle in accordance with 11614318 and associated lists. All parts, subassemblies, and assemblies shall be identified in accordance with MIL-STD-130 when specified on drawings.

3.4.1 Seals.

3.4.1.1 Bearing and lubricant seals. With the vehicle operating under all types of cross-country terrain and during fordings or swimming, the seals shall restrict the entrance of all foreign matter into the bearings, which are exposed to contamination during these operations. Seals shall prevent water contamination of lubricants, and water contamination shall not exceed 2 percent by volume. All bearing seals shall prevent the leaking of lubricants from the bearings.

3.4.2 Special kits. When specified (see 6.2 and 6.4), special kits shall be furnished and meet the requirements specified herein.

3.4.2.1 Rear canopy kit. The rear canopy and frame kit shall conform to items and requirements specified in Drawing 11614938.

3.4.2.2 Shipping and stowage kit. The shipping and stowage kit shall conform to requirements specified in 11632925.

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3.4.2.3 Arctic kit. The arctic kit conforming to 11614988 shall provide a hard top cab for the tractor, heat for the cab, and assure engine starting and operation in ambient temperatures from minus 25°F (4 to 6 feet above ground) to minus 65°F (measured at ground level).

3.4.2.4 Crane arctic kit. The crane arctic kit conforming to 11614987 when used as an operator's station shall operate in temperatures of minus 25°F (4 to 6 feet above ground) to minus 65°F (measured at ground level).

3.4.2.5 Infra-red filter kit. The infra-red filter kit conforming to 11624639 shall be installed.

3.4.2.6 Gun-ring mounting kit. The gun-ring mounting kit conforming to 11614953 when installed shall permit 360 degrees rotation without interference.

3.4.3 Tire chains. The tire chains when installed shall retain maximum tightness at tire manufacturer's rated pressure to prevent damage to wheel walls and vehicle underbody. Scuffing of body metal shall be considered as damage.

3.4.4 Power steering unit. The power steering unit shall operate without malfunction or leakage of lubricants under all vehicle operating conditions.

3.4.5 Engine. The vehicle performance requirements shall be met when engine conforming to MIL-E-62133 is properly installed in vehicle.

3.4.5.1 Engine air induction system. The engine air induction system as fabricated and properly installed from air cleaner to engine turbo charger intake, shall be free from all weld slag and foreign material. The system shall prevent entrance of all foreign material and withstand pressure up to 5 psi. Air cleaner restriction indicator conforming to MS-53063-5 shall be installed.

3.4.6 Transmission. The vehicle performance requirements shall be met when transmission conforming to MIL-T-62134 is properly installed in vehicle.

3.4.7 Exhaust system. The exhaust system as installed shall be gastight and leakproof such that during normal vehicle operation the concentration of NO_x in the crew compartment shall not exceed 5ppm and the concentration of carbon monoxide shall not exceed 50ppm.

3.4.8 Fuel tanks and hydraulic reservoir. Provisions shall be made to assure the internal cleanliness of the fuel tanks and hydraulic reservoir prior to initial filling.

3.4.9 Engine stop switch. When the engine stop switch is activated, the engine shall stop within 2 minutes.

3.4.10 Fuel, air, and hydraulic lines, and fittings. All fuel air, hydraulic lines, and fittings shall be internally clean prior to and after making

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connections and shall be free from leaks. All lines and fittings shall be secured in such a manner to prevent rubbing on adjacent lines or vehicle appendages.

3.4.11 Governor control linkage. Governor control linkage (throttle) shall be adjusted to permit the full range of engine-governed RPM capability. The accelerator pedal travel, when fully depressed, shall be restricted by adjustment to preclude damage to governor and governor control linkage.

3.4.12 Personnel heater (hot water type). The personnel heater MS51326-1, as installed, shall maintain temperatures of not less than 40°F within the vehicle cab in ambient air temperatures down to minus 25°F and provide defrosting of the windshield.

3.4.13 Crane controls. Each control for operation of the crane shall automatically return to neutral position should the operator inadvertently or intentionally release the control. The automatic safety braking for the hoist winch and the check valve in the boom-cylinder hydraulic circuit shall prevent the load from lowering in the event of a power failure. The controls shall permit traverse and lowering of the boom by manual means with and without load in case of power failure and shall permit removal of the load safely and without damage.

3.4.13.1 Crane operators station. Unless otherwise specified (see 6.2), the crane operator's station shall be open with collapsible top without side curtains or windshield.

3.4.13.2 Lifting capacities. The lifting capacities of the crane as installed on the vehicle and while being traversed shall be as follows for the radii specified; or in inverse proportion to load values for radii not shown (with outriggers in place):

(a) At 16 foot radius not less than 10,000 pounds.
(b) At 11 foot radius not less than 15,000 pounds.
(c) At 6 foot radius not less than 20,000 pounds.

3.4.14 Winches.

3.4.14.1 Rear winch. The recovery winch mounted to the rear and within the wrecker body shall have a capacity of not less than 45,000 pounds line pull on the first layer of wire rope and shall be hydraulically controlled in conjunction with or separately from the crane operation.

3.4.14.2 Front winch. The front mounted winch as installed shall have a single line capability of not less than 10,000 pounds on the bare drum. The wire rope for the winch shall be not less than 150 feet in continuous length, without splice or joint and shall be provided with a chain and hook.

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3.4.15 Controls. Electrical, mechanical, pneumatic and hydraulic controls shall operate without malfunction throughout all ranges of operation under all vehicle operating conditions.

3.4.16 Adjustment mechanisms. Adjustment mechanisms shall function properly and maintain adjusted settings during all conditions of vehicle operations.

3.4.17 Stowage. Stowage provision made shall have the capability to contain the manufacturer's, depot and troop installed Equipment List (EL). Padlock as required shall conform to Class II of MIL-P-17802.

3.4.18 Welding repairs and burning operations. Welding repairs and burning operations of any type or class shall be made only when, and to the extent specifically authorized by procuring activity.

3.4.19 Transportability. The vehicle shall be transportable by ship and landing vessels and transportable by railways without major disassembly. Air transportability shall be for C 5A aircraft and shall meet requirement of MIL-A-8421.

3.4.20 Tires. Vehicle equipped with tires conforming to MIL-T-62129 shall meet all requirements specified herein.

3.4.21 Soft top. Unless otherwise specified (see 6.2), the contractor shall furnish the soft top enclosure for the crew compartment.

3.4.22 Safety. The vehicle shall be evaluated in accordance with MIL-STD-882 to assure all hazards have been eliminated or reduced to an acceptable level through the utilization of appropriate safety devices.

3.5 Break-in run. Prior to break-in run, each vehicle produced shall be completely assembled and all adjustments completed (except for equipment, component installation and servicing performed by the contractor as a final processing of vehicles for acceptance). Vehicle, including all components requiring lubrication, shall be lubricated. Vehicle shall be serviced with fuel conforming to VV-F-800. Lubricants and fluids shall be as specified in lubrication chart. Specific applications by grades or types for various expected seasonal temperature ranges shall be as specified therein.

3.5.1 Oil pressure. Prior to start of the break-in run, the engine shall be operated at 700 to 800 RPM until lubricating oil is at operating pressure in accordance with MIL-E-62133. If at any time during the break-in run, the engine has been stopped for at least 15 minutes, the engine shall again be operated as above before continuing the break-in run. Proper oil pressure shall be maintained during operation on level ground, 60% grades and 30% side slopes.

3.5.2 Distance. Each vehicle shall be operated by the contractor for a distance of 15 miles consisting of the following types of operation:

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(a) 1st gear, 1/2 mile
(b) 2nd gear, 1 mile
(c) 3rd gear, 1 1/2 miles
(d) 4th gear, 1 1/2 miles
(e) 5th gear, 4 1/2 miles
(f) 6th gear, 6 miles

NOTE: During the above run, the vehicle shall be stopped and then operated in reverse gear 3 times, 100 feet each time.

3.5.2.1 Condition after run. After completion of the break-in run, the engine shall run smoothly. Prior to vehicle being submitted to the Government for acceptance, the vehicle shall be checked to determine if adjustments are correct. There shall be no defects indicative of damaged components that may cause faulty vehicle operation.

3.6 Performance. The complete vehicle, serviced and equipped for existing environmental and climatic condition and carrying a payload of 8,000 pounds, shall meet performance requirements specified herein.

3.6.1 Environmental. The vehicle shall operate in ambient air temperature of plus 125°F (ground surface temperature) to minus 250°F (4 to 6 feet above grgund) and with special equipment installed, from temperature ranges of minus 25°F (4 to 6 feet above ground) to minus 65°F (ground surface temperature). The complete vehicle when in storage shall withstand climatic extremes as specified in MIL-STD-210 without deterioration that may cause failure of any component part of vehicle.

3.6.2 Grade and slope operation. The vehicle with 8,000 pound payload (without towed load) shall operate in forward and reverse gears on logitudinal grades up to 60% and side slopes up to 30% with each side of the vehicle up slope. As a result of the operation, no evidence of faulty lubrication, cooling, fuel supply, leakage, or other malfunction shall be found.

3.6.3 Engine starting on grades and side slopes. With the vehicle standing on a 60% grade, for not less than 2 minutes with engine operating under no load between 800 to 1000 rpm, the engine shall be stopped for no less than 2 minutes. The engine shall restart in not more than 2 minutes when vehicle is headed up and headed down grade respectively. A similar engine starting test shall be conducted on 30% side slopes with right side and then left side of vehicle up slope.

3.6.4 Cooling system. The engine coolant system and engine and transmission oil coolers as installed shall control the coolant and oil temperatures to prevent damage to the engine and transmissions under all conditions of vehicle operation. Temperatures shall not exceed those shown in Table I when vehicle is operated in ambient temperatures of / 125°F.

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Table I. Operating temperatures.

Maximum engine coolant temperature at engine outlet.	Maximum engine oil temperature measured at oil sump drain plug.	Maximum transmission oil temperature measured at converter outlet.
230°F.	250°F.	275°F.

3.6.5 Cruising range. Vehicle with payload shall operate for not less than 300 miles at average speed of 20 miles per hour on hard-surfaced roads over average rolling terrain without refueling.

3.6.6 Speeds. With 8,000 pound payload, but less towed load; the vehicle shall maintain a speed of not less than 30 mph on level hard-surfaced roads with a grade not greater than ± 1/2 percent, and in low gear maintain a minimum of 2 1/2 miles per hour.

3.6.6.1 Towed loads. The vehicle loaded with its 8,000 pound rated payload shall be capable of towing a disabled fully loaded wrecker vehicle having a gross vehicle weight (GVW) of 46,530 pounds or the fully loaded cargo or tanker vehicle (of lesser GVW). During towed load operations, the vehicle, with payload, shall be operated at not less than 20 mph on a level, hard-surface road. During lift and tow operation, the vehicle shall operate on level hard-surfaced roads at not less than 5 mph on a grade not greater than ± 1/2 percent.

3.6.7 Braking ability.

3.6.7.1 Service brakes. When loaded with rated payload, the service brakes shall stop vehicle from a speed of 20 mph in not more than 35 feet from point of brake application. Deceleration and stopping distance shall be demonstrated on dry, brushed, concrete level road surface, free from loose material with a grade not greater than ± 1/2 percent. The service brake shall hold vehicle on a 60 percent grade when ascending and descending.

3.6.7.2 Parking brake. The parking brake shall hold vehicle loaded with rated payload headed up and headed down on grades of not less than 40 percent.

3.6.8 Steering angle. The power steering mechanism shall be adjusted to permit a turning angle of 60 degrees ± 2 degrees for both left and right turns. These turns shall be made with maximum articulation (roll) between the front and rear units, to either side without interference.

3.6.9 Vehicle articulation. Lateral articulation between the front and rear units shall be 20 degrees ± 2 degrees on each side of the longitudinal centerline. Stops, when used, shall limit the articulation of 5 degrees ± 3

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degrees between the power unit and trailing unit to improve stability during lift and tow operations and during loading for shipment of vehicle by sea, land, and air.

3.6.10 Leakage during swimming. The vehicle with and without rated payload shall swim for a period of not less than 20 minutes. Leakage of water into new vehicle shall not exceed 1 quart per minute, total in tractor and carrier combined.

3.6.10.1 Bilge pumps. The power unit and trailing unit bilge pumps shall remove water from their specific areas at a minimum pumping rate of 46 gpm as installed.

3.6.11 Electrical system.

3.6.11.1 Voltage alternator-generator. With the alternator operating at 2,000 rpm with 25 ampere load, the voltage shall be nominally set at 28 volts. The voltage spread shall not exceed ± .5 volt under all vehicle operating conditions, providing the electrical load does not exceed alternator capacity of 60 amps. The voltage shall be measured at the point of battery connection to the vehicle electrical system.

3.6.11.2 Lights and accessories. The internal and external lights, including cables and receptacles, shall function throughout all vehicle operating conditions. All electrical contacts and connections shall maintain positive contact under all vehicle operating conditions.

3.6.12 Crane performance. The crane shall traverse through 360 degrees at a speed range of 3.5 to 4 rpm's. Controls shall automatically prevent the crane from swinging over the front of the vehicle when the boom is too low to clear the cab. When hoisting a 6,000 pound load, the hook speed shall be 0 to 30 feet per minute. The crane shall have the capability to hoist, traverse, elevate boom simulataneously. The crane shall hold a 5 ton (load carried at rear of vehicle) clear of ground during vehicle operation. The crane equipment shall move the load without any jerky action in all directions in which crane is operated.

3.7 Radio interference suppression. The complete vehicle shall be radio interference suppressed in accordance with tactical vehicle requirements Class IIIA of MIL-STD-461.

3.8 Painting, marking and data plates.

3.8.1 Painting. The exterior and interior of the vehicle, components, assemblies, subassemblies, and parts that require painting shall be prepared and painted in accordance with MIL-STD-193. Vehicle exterior paint and exterior nonskid paint shall match color chip X24087 of FED. STD. No. 595.

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3.8.2 Marking. Marking of vehicle shall be in accordance with drawings and shall conform to MIL-STD-642.

3.8.3 Name, shipping, and identification data plates. All plates shall be in accordance with drawings and shall conform to MIL-P-514.

3.9 Servicing and adjustment. Unless otherwise specified (see 6.2), the contractor shall service the vehicle for operational use prior to delivery. Such servicing shall include, as a minimum, the focusing of lights, the proper adjustment of the engine, running gear, electrical system, brake system, and tire pressure. The engine and all running gear shall be serviced with lubricants of the proper grade for the climatic confitions at the delivery point. Engine cooling system shall be serviced with a solution of ethylene glycol conforming to type I of MIL-A-46153, and water in equal parts of volume.

3.10 Noise and vibration levels. The noise and vibration levels associated with the vehicle and its elements in required combinationa, shall be controlled under all conditions and usage, to levels of tolerance of personnel in accordance with MIL-STD-1472 and MIL-STD-1474.

3.11 Workmanship. The vehicle shall be constructed and finished in a thorough workmanlike manner. Particular attention shall be paid to neatness and thoroughness of marking the parts, plating, painting, machine-screw assemblage, welding and brazing, and freedom of parts from burrs and sharp edges. Electric, hydraulic and lube lines shall be located in a manner so as to prevent possible damage by rubbing on adjacent lines or appendages. No condition shall exist which may create a safety hazard to operating or maintenance personnel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's inspection or quality control system. The contractor shall provide and maintain an effective inspection or quality assurance system acceptable to the Government covering the supplies under the contract. A current, written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor shall notify the Government of any subsequent change to the submitted procedures prior to implementing the change into the system.

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4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied, and (b) Government inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures or instances of poor practices which might have an effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct defects discovered shall be cause for suspension of acceptance until corrective action has been taken, or until conformance of product to specified criteria has been demonstrated.

4.1.3 Materials and qualified products. The contractor shall provide the Government with evidence that materials (see 3.3, 3.3.1, and 3.3.2) used are as specified in the applicable specifications and drawings.

4.1.4 Parts and components. Parts, components and assemblies shall be inspected in accordance with TDPL-11614318.

4.2 Preproduction and initial production inspection.

4.2.1 Preproduction inspection. A preproduction vehicle (see 3.1), when furnished, shall be inspected to ascertain vehicle conformance with the requirements of this specification. Inspection shall be performed at a location approved by the Government. Failure of the preproduction model(s) to comply with any of the requirements specified shall render it unacceptable to the Government until corrective measures satisfactory to the Government have been taken, and a successful retest has been conducted.

4.2.2 Initial production vehicle inspection.

4.2.2.1 Reliability test. Reliability requirement of 3.2.1 shall be verified while the vehicle is subjected to production (comparison) tests. The Minimum Acceptable Value (MAV) shall not be less than 725 Mean Miles Between Mission Failures (MMBMF). This value will be demonstrated at a level of confidence appropriate to the duration of the test.

4.2.2.2 Durability test. Durability requirement of 3.2.2 will be verified while vehicle is subjected to initial production tests.

4.2.2.3 Maintainability. The maintainability of the vehicle as specified in 3.2.3 will be verified during initial production testing.

4.2.2.4 In-process inspection. During fabrication of the initial production vehicle (see 3.2), an in-process inspection will be conducted by a Government representative to evaluate conformance of materials and workmanship to

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requirements of specified documents. These inspections shall be made at the contractor's facility prior to the application of primer and paint. Processing, and welding procedures, quality system and inspection records will be reviewed and evaluated during this inspection.

4.2.2.5 Completed first vehicle inspection. The first production vehicle on each contract shall be road tested and completely inspected by the contractor for conformance to the requirements of contract and specifications. The applicable road test shall be in accordance with Paragraph 4.4.2.1. Upon completion, the contractor shall submit the vehicle (and make available all inspection records and certifications) to the responsible government inspection element at contractor's plant for preliminary inspection.

4.2.2.5.1 Preliminary inspection. The vehicle shall receive a preliminary inspection by the responsible government inspection element. The preliminary inspection shall be completed except for road test requirements of paragraph 4.4.2.1 which will be conducted jointly during provisional inspection to be accomplished by representatives of USATACOM Quality Assurance Directorate as specified in Paragraph 4.2.2.5.2.

4.2.2.5.2 Provisional inspection. The first completed production vehicle shall be subject to provisional inspection at the contractor's plant by representatives of USATACOM Quality Assurance Directorate. The contractor shall further provide any other inspection assistance as may be required. At the time of this inspection, the contractor shall make available his inspection plan, inspection records, and certifications pertinent to the vehicle and components.

4.2.2.6 Repairs of defects. Defects found as a result of above inspections shall be corrected by the contractor at no cost to the Government. Failure of the contractor to promptly correct defects shall be cause for suspension of acceptance of vehicles until corrective action has been accomplished.

4.2.2.7 Vehicle disposition. Upon completion of inspection and acceptance of the preproduction or first production vehicle, the vehicle shall remain at the manufacturing facility as a production sample. The contractor shall service and maintain the vehicle during this period in accordance with the applicable documents for care and preservation while in storage. At the discretion of the Purchasing Contracting Officer, the vehicle may be released for shipment.

4.2.2.8 Initial production test. Unless otherwise specified (see 6.2), to determine conformance to 3.2 through 3.2.3, the quality of vehicles specified in contract will be selected by the Government and shall be subjected to examinations specified in Table III, in addition to road tests specified in Table II and all tests specified in T ble IV. The contractor shall support the test by providing any vehicle spare parts necessary for completion of such a test. Delay in test completion caused by vehicle breakdown or failure of the contractor to adequately support the vehicle with parts during the tests shall note the basis for adjustment of the contract delivery schedule or contract price.

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Table II. 12,500 Mile Test

Course	Mileage & speeds	Vehicle payload (see 3.6)
Hard surface	2,500 miles paved at speeds varying up to maximum	Payload
Secondary roads	5,000 miles at speeds applicable to conditions of terrain	Payload
Hilly cross-country	2,500 miles at speeds applicable to conditions of terrain	Payload
Level cross-country	2,500 miles at speeds applicable to conditions of terrain	Payload
Swimming operation	5 hours of swimming in calm, inland waters	Payload

NOTE: At each 500 mile increments, the crane and winches shall be operated as defined in the mission statement (see 6.5). Ten percent of mileage specified in Table II for each course shall be performed with towed load (see 3.6.6.1) swimming excepted.

4.2.2.8.1 Test failure. Failure of the tes vehicle to meet minimum performance requirements, failure to comply with the requiremeents of a workmanship or materials nature and failure to manufacture the vehicle in accordance with the technical data revealed during or as a result of this test, shall be cause for rejection of the vehicle. Further, the Government may refuse to continue acceptance of production vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate the defects. Any defects found during or as a result of this test shall be prima facie evidence that all vehicles already accepted prior to completion of test, are similarly defective unless evidence satisfactory to the Government is furnished by the contractor that they are not similarly deficient. Such defects on all vehicles shall be corrected by the contractor at no cost to the Government.

4.3 Inspection provisions.

4.3.1 Lot size. An inspection lot shall consist of all vehicles of one type from one day's production, submitted at one time for acceptance inspection.

4.3.1.1 Sampling for inspection. For the purpose of visual, dimensional and primary functional inspection, a representative sample shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105.

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Before sampling may be accomplished by the contractor, the contractor shall 100% inspect the first 20 vehicles to establish a process average (see 6.7) to allow normal sampling in accordance with MIL-STD-105.

4.3.1.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to the applicable drawings and this specification. Examination shall be performed against the classification of defects and with the acceptable quality levels shown in Table III. The following constitutes a part of the classification of defects (see 6.3 and 6.8).

(a) Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.
(b) Any droplet that occurs at a static fit metal to metal or gasket combination shall constitute a major defect.

4.3.1.3 Unclassified defects. All defects having no bearing on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in 5 consecutive lots or ten lots or more within a thirty-day period will be added to the minor classification with no increase to A.Q.L.'s.

4.3.1.4 Recurring major deficiencies. A major deficiency (see 6.8) is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected when found.

Table III. Classification of Quality Defects

Defect No.	Defects	Method of Inspection
Major	A.Q.L. = 25 Defects/100 Units	
101	Steering mechanism: malfunction; unusual noise leaks (see 6.3)	Visual and functional
102	Engine: malfunction; unusual noise; leaks mounting improper	Visual and functional
103	Clutch: malfunction; unusual noise	Visual and functional
104	Transmission: malfunction; unusual noise; leaks (see 6.3) mounting improper	Visual and functional

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Table III. Classification of Quality Defects (Cont'd)

Defects No.	Defects	Method of Inspection
105	Drive train: malfunctional; misalignment, unusual noise	Visual and functional
106	Transfer case assembly: malfunction; unusual noise; leaks, (see 6.3) mounting improper	Visual and functional
107	Compressed air devices: malfunction; unusual noise; leaks	Visual and functional
108	Bilge pumps: malfunction; unusual noise	Functional
109	Cooling system and components: malfunction; unusual noise; leaks; damage; improper clearance	Visual and functional
110	Governor linkage and engine governor: malfunction; adjustment	Visual and functional
111	Fuel system components: malfunction; damage; leaks (see 6.3)	Visual and functional
112	Instrumentation switches, warning, indicating and safety devices: malfunction; damage	Visual and functional
113	Electrical system and components: malfunction; damaged light connectors	Visual and functional
114	Service, handbrakes, and locks: malfunction; unusual noise; leaks	Visual and functional
115	Windshield: fogged; cracked; broken; folding interference	Visual and functional
116	Tire: damage; not properly inflated	Visual
117	Intervehicle cable and hose: improper lengths; damaged	Visual
118	Lubrication system components: (engine) damage; leaks (see 6.3)	Visual and functional

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Table III. Classification of Quality Defects (Cont'd)

Defects No.	Defects	Method of Inspection
119	Body, sheet metal, hardware, storage compartments, brackets: Improper fits; assembly or installation; defective weldment	Visual
120	Tanks structure and welding defect	Visual
121	Hydraulic system: lines; motors; pumps; malfunction unusual noise; leaks (see 6.3)	Visual and functional
122	Crane (controls): malfunction; adjustment damage	Visual and functional
123	Crane (boom): malfunction; adjustment; operation; damages	Visual and functional
124	Exhaust system components: damage; leaks	Visual
125	Air cleaner: damage	Visual
126	Seat suspension system components: damage; leaks	Visual and functional
127	Controls: malfunction; clearance improper	Visual and functional
128	Winch and wire rope: malfunction; improper size; damage	Visual and functional
129	Walkway coating: application; improper coverage; location	Visual
130	Miscellaneous items or accessories: malfunction; damage	Visual and functional

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Table III. Classification of Quality Defects (Cont'd)

Defect No.	Defects	Method of Inspection
Minor	A.Q.L. = 150 Defects/100 Units	
201	Controls: malfunction, adjustment, damage	Visual and functional
202	Coolant: low or improper mixture	Visual and hydrometer
203	Lubricants: levels and types improper (all units)	Visual
204	Wheels and tires: improper size; type and mounting	Visual
205	Pulleys and fan: misalignment; improper mounting or clearance	Visual
206	Bolts: defective, loose	Visual and functional
207	Wiring or tubing: defective: improper assembly or assembly	Visual
208	Brake system components: improper assembly installation, or protection	Visual
209	Body, doors, access ladder, access covers, ventilators, stowage brackets, tie downs, lashing hooks and lifting devices: improper fits; assembly or installation; defective weldments: bows; flooring; seals, hardware	Visual and functional
210	Sheet metal and welding defects	Visual
211	Electrical system components: improper installation or assembly	Visual
212	Paint: application and color improper	Visual
213	Steering system components: improper installation or clearance	Visual and functional
214	Exhaust system components: improper assembly or installation	Visual

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Table III. Classification of Quality Defects (Cont'd)

Defects No.	Defects	Method of Inspection
Minor	A.Q.L. = 150 Defects/100 Units	
215	Cooling system components: improper assembly or installation	Visual
216	Fuel system components: improper assembly or installation	Visual
217	Lube fittings: defective; missing or improperly installed	Visual
218	Insulation and noise suppression: missing: improper application	Visual
219	Mud flaps: improper assembly; damaged	Visual
220	Walkway coating: excessive application	Visual
221	Seat suspension system components: improper installation or assembly	Visual
222	Cushions, seats, straps, canvas tops, and curtains: damaged; improper fit and installation	Visual
223	Miscellaneous items or accessories: missing; improper assembly or installation	Visual and functional
224	Decals, marking, data and instruction plates: incomplete data; missing, improper location or size; vehicle registration and identification marking	Visual
225	Protective coatings: application; missing or improperly processed	Visual
226	Lubrication improper	Visual
227	Tools: missing; improperly processed or secured	Visual

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4.4 Classification of tests.

(a) Acceptance tests, 4.4.1 through 4.4.1.20.
(b) Control tests, 4.4.2 through 4.4.2.9.
(c) Comparison tests, 4.4.3 through 4.4.3.13.

4.4.1 Acceptance test. Each complete vehicle shall be subjected to a 15 mile road test by the contractor over hard-surfaced roads with a grade not greater than ± 1/2 percent. Vehicles (less payload) shall comply to the performance requirements specified herein. Each vehicle shall be operated at speeds specified in 3.5.2 and operated in a reverse gear direction for a minimum of 100 feet 3 times and shall meet all acceptance tests specified at place of manufacture, (see Table IV).

Table IV. Location of tests

	Requirement paragraphs	Test paragraphs	Place of mfr.	Proving Ground check
Preproduction vehicle	3.1	4.2.1	X	
Initial production vehicle inspection	3.2	4.2.2	X	X
Reliability	3.2.1	4.2.2.1		X
Durability	3.2.2	4.2.2.2		X
Maintainability	3.2.3	4.2.2.3		X
In-process (inspection)		4.2.2.4	X	
Complete vehicle inspection		4.2.2.5	X	X
Repair of defects		4.2.2.6	X	
Vehicle disposition		4.2.2.7	X	
Initial production test		4.2.2.8		X
Test failure		4.2.2.8.1		X
Inspection provisions		4.3		
Construction & acceptance tests	3.4	4.4.1	X	X
Power steering unit check	3.4.4	4.4.1.2	X	X

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Table IV. Locations of tests (Cont'd)

	Requirement paragraphs	Test paragraphs	Place of mfr.	Proving Ground check
Engine and engine air induction system check	3.4.5 & 3.4.5.1	4.4.1.3	X	X
Transmission check	3.4.6	4.4.1.4	X	X
Exhaust system check	3.4.7	4.4.1.5	X	X
Fuel tanks, engine stop switch & hydraulic reservoir check	3.4.8 & 3.4.9	4.4.1.6	X	X
Fuel, air & hydraulic lines and fitting check	3.4.10	4.4.1.7	X	X
Governor control linkage	3.4.11	4.4.1.8	X	X
Personnel heater check	3.4.12	4.4.1.9	X	X
Crane, controls, operators station check	3.4.13 & 3.4.13.1	4.4.1.10	X	X
Rear and front winch and cable check	3.4.14, 3.4.14.1 & 3.4.14.2	4.4.1.11	X	X
Controls and adjustment check	3.4.15 & 3.4.16	4.4.1.12	X	X
Welding repairs & burning operations	3.4.18	4.4.1.13	X	X
Tire check	3.4.20	4.4.1.14	X	X
Break-in run & condition after run check	3.5 thru 3.5.2.1	4.4.1.15	X	X
Brake test		4.4.1.16	X	X
Steering and articulation functional test	3.6.8 & 3.6.9	4.4.1.17	X	X

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Table IV. Locations of tests (Cont'd)

	Requirement paragraphs	Test paragraphs	Place of mfr.	Proving Ground check
Leakage during swimming test	3.6.10 & 3.6.10.1	4.4.1.18	X	X
Voltage alternator lights & accessories check	3.6.11.1 & 3.6.11.2	4.4.1.19	X	X
Servicing and adjustment check	3.9	4.4.1.20	X	X
Control tests		4.4.2	X	X
Seals check	3.4.1 & 3.4.1.1	4.4.2.2	X	X
Special kits check	3.4.2 thru 3.4.2.6	4.4.2.3	X	X
Tire chains check	3.4.3	4.4.2.4	X	X
Crane performance test	3.4.13 thru 3.4.13.2 & 3.6.12	4.4.2.5	X	X
Winch test	3.4.14.1 & 3.4.14.2	4.4.2.6	X	X
Soft top check	3.4.21	4.4.2.8	X	X
Vehicle operational test		4.4.2.7	X	X
Grade and slope operation test	3.6.2	4.4.2.7.1	X	X
Engine starting on grades & side slope tests	3.6.3	4.4.2.7.2	X	X
Cooling system test	3.6.4	4.4.2.7.3	X	X
Speed test	3.6.6 & 3.6.6.1	4.4.2.7.4	X	X
Brake test	3.6.7.1 & 3.6.7.2	4.4.2.7.5	X	X

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Table IV. Locations of tests (Cont'd)

	Requirement paragraphs	Test paragraphs	Place of mfr.	Proving Ground check
Holding on grade test	3.6.7.1 & 3.6.7.2	4.4.2.7.6	X	X
Steering angle & vehicle articulation test	3.6.8 & 3.6.9	4.4.2.7.7	X	X
Bilge pump operation	3.6.10.1	4.4.2.7.8	X	X
Comparison tests		4.4.3		X
Special kits operational tests	3.4.2 thru 3.4.2.6	4.4.3.2		X
Carbon monoxide concentration	3.4.7	4.4.3.3		X
Crane performance test	3.4.13 thru 3.4.13.2 & 3.6.12	4.4.3.4		X
Rear and front winch operational test	3.4.14.1 & 3.4.14.2	4.4.3.5		X
Transportability check	3.4.19	4.4.3.6		X
Vehicle performance test	3.6, 3.6.2 & 3.6.5	4.4.3.7		X
Environmental operating conditions test	3.6.1	4.4.3.8		X
Leakage during swimming operational test	3.6.10	4.4.3.9		X
Bilge pump test	3.6.10.1	4.4.3.9.1		X
Radio interference suppression test	3.7	4.4.3.10		X
Noise and vibration level test	3.10	4.4.3.11	X	X
Preservation, packaging, and vehicle processing	5.1	4.5	X	

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4.4.1.2 Power steering unit check. To determine conformance to 3.4.4, the power steering unit shall be checked to assure proper assembly, installation and leaks.

4.4.1.3 Engine and air induction system check. To determine conformance to 3.4.5 and 3.4.5.1, the engine and engine air induction system shall be checked for proper assembly and installation.

4.4.1.4 Transmission check. To determine conformance to 3.4.6, the transmission shall be checked for proper assembly and installation.

4.4.1.5 Exhaust system check. To determine conformance to 3.4.7, the exhaust system shall be checked for proper installation and for leakage.

4.4.1.6 Fuel tanks, hydraulic reservoir, and engine stop switch check. To determine conformance to 3.4.8 and 3.4.9, the fuel tanks and hydraulic reservoir shall be checked for cleanliness, leaks, chaffing, and vibration. The engine stop switch shall be operated and checked for functional requirements.

4.4.1.7 Fuel, air, hydraulic lines, and fittings check. To determine conformance to 3.4.10, the lines and fittings shall be checked for leaks and signs of excessive vibration and chaffing that could lead to vehicle failure.

4.4.1.8 Governor control linkage check. To determine conformance to 3.4.11, the throttle pedal shall be depressed to stop position and the engine governor control linkage checked to assure that full rack position has been reached without binding or damage to linkage or governor.

4.4.1.9 Personnel heater check. To determine conformance to 3.4.12, the heater shall be checked for proper assembly and installation.

4.4.1.10 Crane controls and operators station check. To determine conformance to 3.4.13 and 3.4.13.1, the crane equipment and controls shall be checked for completeness, proper installation and damage. The walking areas and platforms shall be checked for sufficient and proper application of non-skid materials.

4.4.1.11 Rear winch, front winch, and wire rope check. To determine conformance to 3.4.14.1 and 3.4.14.2, the rear and front winch, wire rope, and their components shall be checked for completeness, proper fit, installation and damage.

4.4.1.12 Controls and adjustment mechanisms check. To determine conformance to 3.4.15 and 3.4.16, adjustment mechanisms and all controls and mechanisms shall be operated and checked for functional requirements.

4.4.1.13 Welding repairs and burning check. To determine conformance to 3.4.18, welding repairs and burning procedures shall be checked for use of procedures.

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4.4.1.14 Tire check. To determine conformance to 3.4.20, the tires shall be checked for proper type and pressures.

4.4.1.15 Break-in run and condition after break-in check. To determine conformance to 3.5 through 3.5.2.1, each vehicle less payload, shall be operated as specified and observed for performance requirements and condition after the break-in run.

4.4.1.16 Brake test. The vehicle shall be operated, service and parking brakes checked for meeting functional requirements, except for grade requirements.

4.4.1.17 Steering and articulation functional test. To determine conformance to 3.6.8 and 3.6.9, vehicle shall be operated and functional tests made.

4.4.1.18 Leakage during swimming test. To determine conformance to 3.6.10 and 3.6.10.1, the vehicle shall be placed or driven into water of sufficient depth to float vehicle, leakage shall not exceed that specified. Bilge pumps shall be functionally tested.

4.4.1.19 Voltage alternator, lights and accessories check. To determine conformance to 3.6.11.1 and 3.6.11.2, the alternator voltage shall be checked at point specified and voltage shall be as specified with all lights and accessories functioning.

4.4.1.20 Servicing and adjustment check. To determine conformance to 3.9, the vehicles shall be checked for conformance to requirements specified therein.

4.4.2 Control tests.

4.4.2.1 Frequency. The Government shall select at random for control test one (1) out of each 40 vehicles submitted or a minimum of one per month, if production is less than 40 vehicles per month. Each vehicle selected shall be operated, with designated or simulated payload, for a distance of 50 miles by the contractor. These tests shall be performed in the presence of a Government representative. After completion of tests; the doors, control devices, and other functional parts shall be tested for leaks and functional requirements.

4.4.2.1.1 Failure. If the vehicle selected fails to pass any of the control tests, the Government inspector shall stop acceptance examination and testing on subsequent vehicles until such time as conditions causing the failure have been remedied. Any defects found during or as a result of the test shall be prima facie evidence that vehicles accepted subsequent to the previous acceptable control test vehicle are similarly defective unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similar defective. Such defects, on all vehicles, shall be corrected by the contractor at no cost to the Government. Another vehicle with corrective actions implemented shall be subjected to the control test.

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4.4.2.2 Seals check. To determine conformance to 3.4.1.1, after fording or swimming operations; all bearings subject to water contamination shall be checked. Lubricants shall be drained and examined for water contamination and shall not be more than that specified.

4.4.2.3 Special kits check. To determine conformance to 3.4.2 through 3.4.2.6 one of each 10 samples of a specific kit furnished shall be installed in accordance with applicable drawings.

4.4.2.4 Tire chains check. To determine conformance to 3.4.3, the tire chains shall be installed and vehicle driven for not less than 2 miles, and retention of tightness and clearance in wheel well checked.

4.4.2.5 Crane performance test. To determine conformance to 3.4.13 through 3.4.13.2 and 3.6.12, the crane with specified load shall be operated with the use of outriggers and shall meet performance requirements specified. All overload and safety devices shall be tested for correct operation.

4.4.2.6 Rear and front winch, and wire rope test. To determine conformance to 3.4.14.1 and 3.4.14.2, the winch and rope shall be tested with rated load and observed for meeting requirements.

4.4.2.7 Vehicle operational test. The vehicle loaded with rated payload including all stowage and OEM item (see 3.4.17) shall be tested as specified herein. The testing, where applicable, may be performed during the 50 mile road test. The contractor shall be permitted to unpackage and repackage all stowage and OEM items used for test. Troop installed and depot installed OEM will be furnished by the Government.

4.4.2.7.1 Grade and slope operation test. To determine conformance to 3.6.2, the vehicle shall be operated on 60 percent grades headed up and down and on 20 percent side slopes with each side of vehicle up slope.

4.4.2.7.2 Engine starting on grades and side slopes test. To determine conformance to 3.6.3, the vehicle shall be placed on specified grades and slopes, and engine stopped then restarted to assure engine meets starting requirements. The fuel and lubrication systems shall function in all positions without failure.

4.4.2.7.3 Cooling system test. To determine conformance to 3.6.4 during the road test, the cooling system shall be checked to assure the system meets cooling requirements. The maximum temperatures of the transmission oil and engine coolant shall be recorded. Engine oil pressure shall also be recorded.

4.4.2.7.4 Speed test. To determine conformance to 3.6.6 and 3.6.6.1, the vehicle loaded with rated payload shall meet the speed requirements.

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4.4.2.7.5 Braking test. To determine conformance to 3.6.7.1 and 3.6.7.2, during road test the service and parking brakes shall be checked for functional operation. Stopping distance shall be not more than that specified.

4.4.2.7.6 Holding on grades test. To determine conformance to 3.6.7.1 and 3.6.7.2, the vehicle loaded with rated payload shall be operated on 60 and 40 percent grade as specified, the vehicle shall be stopped and the transmission put in neutral, service and parking brakes shall hold the vehicle on the specified grades.

4.4.2.7.7 Steering angle and vehicle articulation test. To determine conformance with 3.6.8, the vehicle shall be operated over terrain permitting articulation (roll) between the front and rear units. The vehicle shall be turned to check conformance with the requirement specified to the right and the left to assure that there is no binding or interference between the front and trailing units under all conditions of steer and roll. To determine conformance with 3.6.9, stops shall be placed to check the roll limitation as specified.

4.4.2.7.8 Bilge pump operational test. To determine conformance to 3.6.10.1, bilge pumps shall be operated during swimming test to determine performance and rate of water removal.

4.4.2.8 Soft top check. To determine conformance to 3.4.21, the top shall be inspected for completeness of assembly.

4.4.3 Comparison tests. The Government may select vehicles anytime during a production contract period and subject vehicles to all tests listed on Table IV and Table V to reveal defects of a workmanship or materials nature that may reduce the effective operation of these items in the field and compare existing quality with previous standards. These tests shall be conducted at Government laboratories or proving grounds designated by the contracting officer. Comparison test vehicles shall be combat loaded with all OEM stored in their applicable spaces. Vehicles selected shall not include any vehicles previously tested for conformance to 4.4.2 (Control tests).

4.4.3.1 Test failure. Failure of the test vehicle to meet minimum performance requirements, failure to comply with the requirements of a workmanship or materials nature and failure to manufacture the vehicle in accordance with the technical data revealed during or as a result of this test, shall be cause for rejection of the vehicle. Further, the Government may refuse continue acceptance of production vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate the defects. Any defects found during or as a result of this test shall be prima facie evidence that all vehicles already accepted prior to completion of tests, are similarly defective unless evidence satisfactory to the Government is furnished by the contractor that they are not similarly deficient. Such defects on all vehicles shall be corrected by the contractor at no cost to the Government.

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Table V. Comparison test (2500 miles)

Course	Mileage & speeds	Vehicle payload (see 3.6)
Paved surface (20%)	500 miles at speeds varying up maximum	8000 pounds
Secondary roads (36%)	900 miles at speeds applicable to terrain	8000 pounds
Hilly cross-country (20%)	500 miles at speeds applicable to terrain	8000 pounds
Level cross-country (20%)	500 miles at speeds applicable to terrain	8000 pounds
Belgian block (4%)	100 miles	8000 pounds
Swimming	1 hour	8000 pounds

NOTE: (a) At each 500 mile increment, the crane and winch shall be operated as defined in mission statement (see 6.5).
(b) 10% of mileage for each course shall be with towed load. Two miles of lift and tow operation required on level paved surface.

4.4.3.2 Special kits operational tests. To determine conformance to 3.4.2 through 3.4.2.6, when special kits are furnished for testing they shall be installed on vehicle and tested for meeting performance requirements for the individual kit.

4.4.3.3 Exhaust concentration check. To determine conformance to 3.4.7, the occupied areas of the vehicle shall be checked for concentration of exhaust fumes. Fumes shall not exceed percent specified.

4.4.3.4 Crane performance test. To determine conformance to 3.4.13 through 3.4.13.2 and 3.6.12, the crane with specified load shall be checked for lifting capacities using the different radii with outriggers. Checks shall be made to check operation of all overload and safety controls.

4.4.3.5 Rear and front winch operational test. To determine conformance to 3.4.14.1 and 3.4.14.2, the winch is with the specified rope shall be tested by winching rated loads that will check the required line pull and safety overload devices.

4.4.3.6 Transportability check. To determine conformance to 3.4.19, the vehicle shall be checked for transportability requirements.

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4.4.3.7 Vehicle performance tests. To determine conformance to 3.6, the vehicle with rated payload shall meet all performance requirements and test results recorded.

4.4.3.8 Environmental operating conditions tests. To determine conformance to high and low temperature operation of 3.6.1, the vehicle shall be tested as required to assure that the vehicle meets all environmental requirements.

4.4.3.9 Leakage during swimming tests. To determine conformance to 3.6.10, the vehicle shall be operated as specified, performance and leakage checked.

4.4.3.9.1 Bilge pump test. To determine conformance to 3.6.10.1, bilge pumps shall be operated during swimming test to determine performance and rate of water removal.

4.4.3.10 Radio interference suppression test. To determine conformance to 3.7, the vehicle and components shall be subjected to the tactical vehicle suppression requirements of specification specified.

4.4.3.11 Noise and vibration level tests. To determine conformance to 3.10, the applicable sound levels, vibration level, instrumentation shall be used to assure the vehicle meets requirements.

4.5 Preservation, packaging, and vehicle processing. Material and equipment shall be inspected prior to shipment to assure conformance to the applicable requirements specified in 5.1.

5. PREPARATION FOR DELIVERY

5.1 Vehicle processing. Unless otherwise specified, the vehicle and equipment shall be processed for shipment and storage in accordance with MIL-V-62038 to the extent indicated on the applicable vehicle preservation data sheet or as specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The vehicle covered by this specification is intended to support off-road vehicles in the field during tactical military assigned mission such as lifting and towing of disabled vehicles and for replacement of major components of combat vehicles on site under extreme climatic conditions, and transportable in Phase III air lift.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction vehicle, if required (see 3.1).
(c) Special kits, if required (see 3.4.2).

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(d) Crane operators enclosure, if required (see 3.4.13.1).
(e) Quantity of initial production test vehicles, if required (see 4.2.2.8).
(f) Level of preservation, packaging, and processing if other than specified (see 5.1).

6.3 The following definitions shall be used for the classification of defects for leaks:

(a) Weep: - Any non-recurring evidence of fluid beyond the seal.
(b) Seep: - Any recurring evidence of fluid beyond the seal that does not result in formation of a droplet.
(c) Droplet: - Any recurring evidence of fluid beyond the seal that results in formation of a non-falling droplet.
(d) Drip: - Any recurring evidence of fluid beyond the seal where droplet forms and falls.

6.4 Special kits (see 6.2).

11614953 - Gun-ring mounting kit.
11614988 - Arctic kit.
11614987 - Crane arctic kit.
11624639 - Infra-red filter kit.
11614938 - Rear canopy kit.
11632925 - Shipping and stowage kit.

6.5 Mission statement. One mission of the M553 truck wrecker, 10 ton, shall consist of 50 miles of operation under the conditions of load, speed, terrain, and environment shown below and performance of the special functions described.

Terrain	Distance	Operating MPH Range
Hard surface	10 miles	0 - 30
Secondary	20 miles	0 - 30
Level cross-country	8 miles	0 - 20
*Level cross-country (MUD)	2 miles	0 - 8
*Hilly cross-country	10 miles	0 - 12
*Swimming	Traverse Water Barrier	5 Minutes Operation

60% Grades without towed load: 1/10 of 1% of Hilly Cross-Country Operation
30% Slopes without towed load: 1/10 of 1% of Hilly Cross-Country Operation

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CLIMATIC EXTREMES	
Temperature:	-25° F to 125° F ambient
Altitude:	500' to 5,000'

TRACTOR FUNCTIONS	
Idle:	30 minutes
Starts:	One cold, one hot at two minutes, and hot at 15 minutes after shutdown. (Cold means coolant and oil at ambient, hot at maximum obtained)
*4-Wheel Drive:	100% Mud and Swim, cross-country as required, all other operation is in 2-wheel drive
Reverse:	10 operations at 130 feet each

RECOVERY SYSTEM FUNCTIONS:

Rear winch: (One each 10 missions) A tow winch recovery operation, with not less than 45,000 pounds line pull on the first layer of wire rope, for a distance of 200 feet.

Front winch: (One each 10 missions) A winch recovery operation, with not less than 10,000 pounds on the bare drum.

Lift and tow: An equivalent GVW vehicle lift and tow operations (once each 5 missions) over level, hard, surface roads for two miles.

Crane: The crane shall be operated (once each 5 missions) under the following conditions:

(1) (One cycle) Boom horizontal, no load, rotate clockwise and counter-clockwise, activating automatic stop microswitches at 135° right and left from boom travel position.

(2) (One cycle) With upper boom fully extended and at a 8 foot radius, activate safety overload pressure sensing device by attempting to lift a 20,000 pound load.

(3) (One cycle) With upper boom fully retracted and at a 6 foot radius, traverse 360° clockwise counter-clockwise a 20,000 pound load.

(4) (Two cycle) With upper boom fully extended and retracted at a 16 foot radius, traverse 360° clockwise and counter-clockwise a 10,000 pound load.

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6.6 Typical vehicle life operational profile.

Model	Renewal Life Miles - Hours	Avg M.P.H.	Total Missions	Mission Miles - Hrs
M520	25,000 - 1250	20.0	250	100 - 5
M559	18,000 - 1250	14.4	182	100 - 6.9
M553	12,500 - 1250	10.0	250	50 - 5
M877	25,000 - 1250	20.0	250	100 - 5

In actual troop use, this will vary due to environmental, terrain, and troop mission differences.

6.7 Process average. Sampling may be initiated if the process average value of the first twenty vehicles inspected is less than the A.Q.L. specified in the classification of defects for major and minor defects. Process average shall be computed as follows:

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of vehicles inspected}} \times 100$$

If the computed process average exceeds the specified A.Q.L., 100 percent inspection shall be performed and continued until such time that the process average for twenty consecutive vehicles is less than the specified A.Q.L.

6.8 Definitions of recurring major and minor defects. The Government Inspector will verify that a thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but also for any other departures from good workmanship. The Government Inspector will assure that all deficiencies encountered during the inspection are enumerated on the Deficiency Sheet for the vehicle. The defects noted on Deficiency Sheet shall contain sufficient description to enable the Government Inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the vehicle specification and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. 2320-A956

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-T-62131B(AT)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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